Journal of the American Society for Mass Spectrometry

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Perspective

Developing Single-Cell Proteomics to Open New Perspectives on Biology

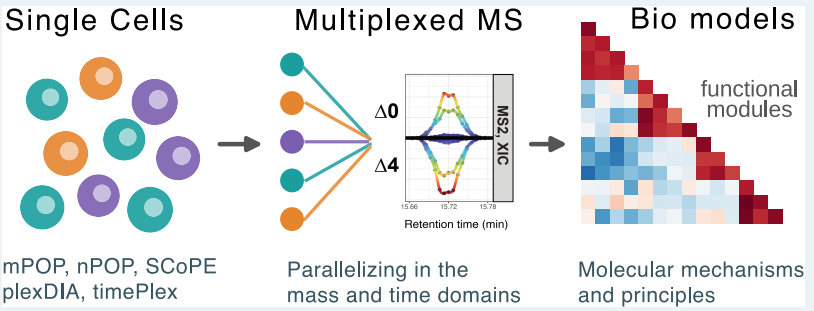
Published as part of Journal of the American Society for Mass Spectrometry special issue "Single Cell Proteomics: Honoring Nikolai Slavov, Recipient of the 2026 ASMS Biemann Medal".

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Cite This: https://doi.org/10.1021/jasms.6c00362

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ABSTRACT: Mass-spectrometry-based single-cell proteomics has evolved from an unconventional idea into a rapidly advancing technology that is opening new perspectives on biology. In this perspective, I reflect on the development of the field over the past decade: the guiding ideas, the early experiments that established feasibility, the advances in sample preparation, mass spectrometry methods, instrumentation, computation, and multiplexing that transformed sensitivity and throughput, and the collaborative efforts that made these technologies increasingly reproducible and accessible. I highlight biological results from the first decade and questions that have become addressable with highly quantitative protein measurements in individual cells, including functional protein coordination, regulation of protein synthesis and degradation, developmental decisions, and disease progression. Looking forward, I foresee that continued technological innovation, rigorous benchmarking, and an open and critically engaged community will be essential for realizing the broader promise of single-cell proteomics: turning high resolution protein analysis into a powerful source of biological insight, including generalizable principles.

INTRODUCTION

Receiving the Biemann Medal has given me an occasion to look back on a decade of research, much of it collaborative, that grew into what is now called mass-spectrometry-based single-cell proteomics. This account is not a comprehensive review of the field; several comprehensive ones already exist.[1,2] It is instead a personal narrative of how the technology came to be, the people who built it alongside me, key biological results focused on new perspectives opened by the technology, and the reasons I believe it is essential to biology going forward.

MOTIVATION FOR SINGLE-CELL ANALYSIS

My interest in single-cell analysis began when I was a PhD student at Princeton University studying a puzzling phenomenon: synchronized cultures of yeast cells exhibiting oscillatory oxygen consumption. We hypothesized that this population-level oscillation reflected a cell-intrinsic cycle present even in unsynchronized cultures and that it can explain their growth rate responses.[3−5] I was also fascinated by a simple mechanism for decoding the frequency of calcium oscillations into differential regulation of target proteins.[6] The most direct and rigorous way to test our hypotheses about the functions of these cellular oscillations was to analyze single cells. At the time, that meant fluorescence in situ hybridization: imaging cells hybridized with fluorescent nucleic acid probes and writing scripts to count individual mRNA molecules.[7]

These experiments helped me appreciate both the power and the limitations of single-cell analysis restricted to RNA, particularly the difficulty of counting low-copy-number molecules reliably even when using single-molecule resolution

Received: August 25, 2026
Revised: September 27, 2026
Accepted: October 19, 2026

ACS Publications

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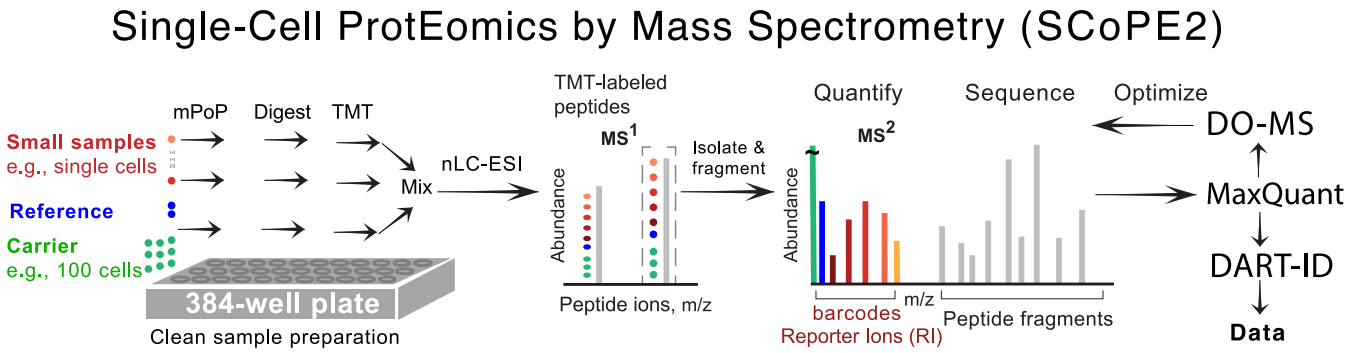


Figure 1. Schematic representation of the second-generation SCoPE-MS, SCoPE2. It shows the automated sample preparation in multiwell plates, the use of an isobaric carrier, and the computational tools used for data interpretation and optimization. Abbreviations: SCoPE2—Single Cell ProtEomics by Mass Spectrometry version 2; DO-MS—Data-Driven Optimization of Mass Spectrometry; DART-ID—Data-driven Alignment of Retention Times for Identification;[23] mPOP—Minimal ProteOmic sample Preparation;[24,25] The figure is adapted from ref 26. Copyright 2020 Nikolai Slavov.

methods, and I began thinking about extending single-cell analysis to proteins. While mass spectrometry (MS) analysis had detected individual highly abundant peptides and proteins in single cells, these cases were generally considered exceptions (reviewed in refs[8−10]) that did not support the feasibility of scalable proteomic analysis of typical mammalian cells.[11,12]

OUR FIRST SINGLE-CELL PROTEOMIC EXPERIMENTS

When I started my independent position as an assistant professor at Northeastern University in 2015, I had the opportunity to begin developing protein measurements in single mammalian cells using MS. At the time, this was an unorthodox idea that many senior colleagues considered infeasible. Yet, I believed it was feasible based on my estimates of protein copy numbers and MS instrumentation capabilities. A well-known fact gave me considerable encouragement: most proteins in human cells are present at many thousands of copies per cell.[13] If we could develop methods that did not lose the majority of the protein material during sample preparation, existing mass spectrometry instrumentation should already have sufficient sensitivity to quantify—not merely detect—many proteins per cell.[14] The distinction between detecting a protein and quantifying it reliably, with accuracy sufficient to support biological inference, has remained a guiding principle throughout the research of my teams.

To demonstrate the feasibility of single-cell proteomics by mass spectrometry, I needed a good team and access to MS instruments. My Northeastern startup was modest and did not include an allowance for purchasing a mass spectrometer. Thus, our early single-cell proteomics work could not rely on an instrument in my laboratory. Fortunately, I had a good friend, Dr. Bogdan Budnik, who managed the FAS Harvard mass spectrometry facility, and our collaboration provided indispensable access to MS instruments. To complement the team, I recruited Ezra Levy, who at the time was an undergraduate at Northeastern University. He was bright and motivated by the challenge, perhaps unaware that many senior colleagues considered the project essentially impossible.

USING AN ISOBARIC CARRIER

A key component of our success was the realization that the central bottleneck was not simply detecting a peptide-specific precursor ion, but acquiring enough fragment ions to achieve confident, specific peptide identification from the small amount of material in a single cell.[15] To address that bottleneck, we developed an isobaric carrier strategy, now generally known by the name of the method it enabled, Single Cell ProtEomics by Mass Spectrometry (SCoPE-MS).[16] Using tandem mass tag (TMT) reagents, we labeled multiple individual cells, each with a distinct label, alongside a small bulk "carrier" sample. Because the fragment ion signal is the cumulative product of peptide molecules across all labeled channels, including the carrier, this design substantially increased the probability of confidently identifying peptides from single cells while preserving independent quantification per cell. The isobaric carrier also helped mitigate nonspecific losses of peptides due to surface adhesion.

Our first experiments were noisy, but the signal was unmistakable: We could reliably quantify protein levels that distinguished single cells from different cell types. Crucially, we could confirm that these protein differences corresponded to the cell-type-specific proteomes measured by conventional analysis of larger samples from the corresponding cell types. We had a limited budget, instrument time, and equipment, and all these limitations imposed natural constraints on the experiments that we could perform. However, I did not find these constraints discouraging; rather, knowing how suboptimal our implementation still was made the results more encouraging because we appreciated the enormous headroom to improve.

We first shared our results in a preprint published in January 2017.[16] It immediately attracted much attention, excitement, and skepticism. Our results deserved scrutiny—we were using comparatively old instrumentation to claim measurements that many considered out of reach even for cutting-edge technology. This prompted some colleagues to reflexively reject our results without in-depth evaluation and delayed the acceptance of our article by peer reviewers. It was rejected by several journals and was finally published in *Genome Biology*,[17] by which point the data had been reevaluated by the community and the methodology increasingly accepted. This happened because some colleagues, including senior leaders in the field, read, questioned, and evaluated our results rather than dismissing them outright. This culture of engaged skepticism, rather than reflexive rejection, was essential to the field's development. I remain very grateful for that response of curious skepticism and thoughtful, critical evaluation.

The acceptance of our provocative results was substantially facilitated by making our data easily accessible to examine and reevaluate.[18] Furthermore, the experimental design was relatively simple and straightforward for colleagues to repeat,

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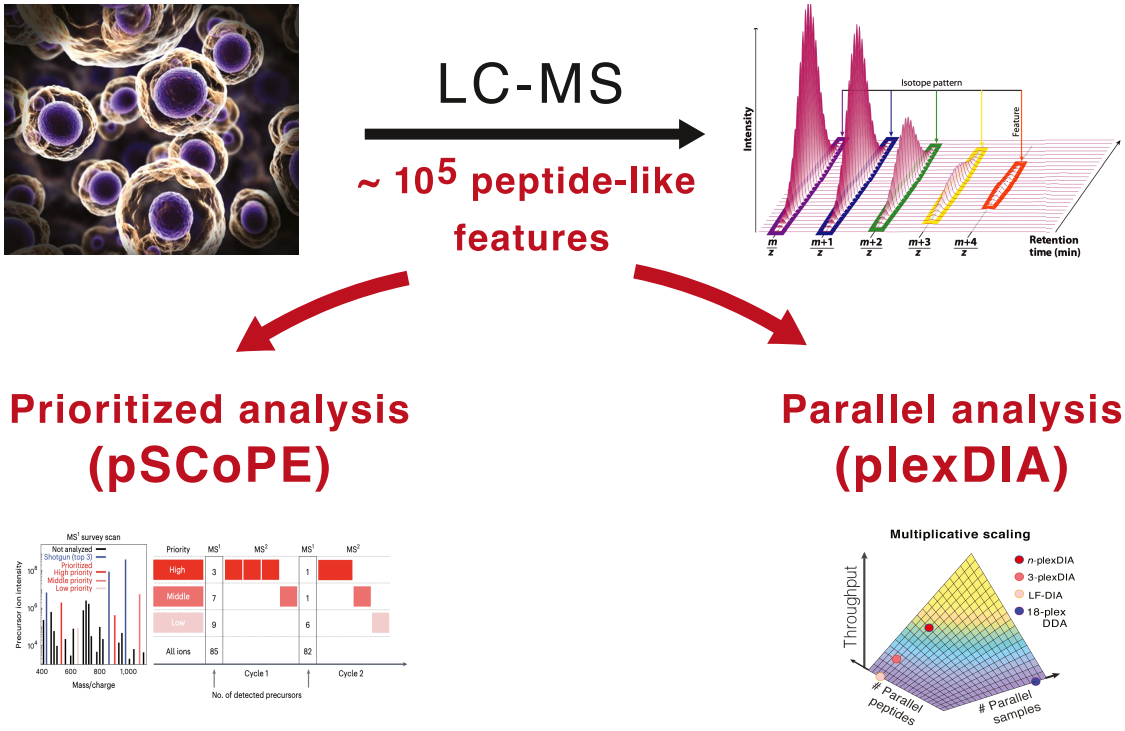


Figure 2. Strategies to increase the fraction of peptide-like features with confidently assigned sequences. While LC-MS analysis of a single cell can detect many peptide-like features, only a small fraction of them can be isolated individually and sequentially for fragmentation, especially since their low abundance requires relatively long accumulation times for productive MS2 scans.[44] Strategies that can increase confident sequence assignment include prioritized data acquisition[48] and parallelized data acquisition by DIA and plexDIA.[50] Abbreviations: pSCoPE—Prioritized Single Cell ProtEomics by Mass Spectrometry; plexDIA—Multiplexed Data Independent Acquisition.

reproduce, and evaluate for themselves[9,19,20] and continues to support biological analysis and discoveries.[21,22] The independent replications of our methodology strengthened the evidence for, and credibility of, the early single-cell proteomics results. Gradually, the skepticism began shifting toward enthusiasm for the capabilities of mass spectrometry proteomics to quantify many proteins in single mammalian cells.[11,12] This transition was enabled by the contributions of many colleagues whose results cumulatively made the field credible, as discussed in the section Building a Community and Outlook.

Thanks to support from the NIH Director's New Innovator Award, my laboratory acquired a Q Exactive mass spectrometer, which helped us to substantially improve sample preparation and data acquisition protocols, resulting in a second-generation method (SCoPE2)[24,25,27] shown in Figure 1. Having an instrument in our laboratory allowed us to perform many more experiments, and we used this to focus on careful characterization of bottlenecks in our workflows through a series of well-controlled experiments. We developed the Data-Driven Optimization of Mass Spectrometry (DO-MS) framework as a set of diagnostic plots reflective of performance at each step of mass spectrometry data acquisition.[28] It aims to diagnose as specifically as possible what the challenges in any one experiment are and therefore facilitate troubleshooting and optimizing data acquisition.

SCoPE2 introduced automated MS-compatible sample preparation by mPOP in multiwell plates, which substantially increased the number of single cells that could be prepared in parallel.[24] At the same time, optimized data acquisition with shorter chromatographic gradients substantially increased the rate at which single cells could be analyzed, and improved sensitivity increased the depth of proteome coverage per cell. Because SCoPE2 relied on commercially available reagents, standard instrumentation, and a detailed, openly shared protocol,[25] it was readily adopted by the wider community and applied to diverse biological systems. This accessibility gave a significant thrust to the development of the field.

All of this ushered in a period of steady growth and improvement in our methodology and generation of data that supported biological inferences with increasing confidence. We felt confident enough to use our technology to answer challenging, long-standing questions such as the molecular mechanisms of macrophage polarization[27] and symmetry breaking in early mammalian development.[29]

SAMPLE PREPARATION

The importance of sample preparation was obvious from the very beginning. The ability to maximize the efficiency of transferring proteins from single cells to MS detectors and to minimize contamination was essential. The nanoPOTS platform developed by the Kelly group was among the first to demonstrate that sample preparation could be miniaturized to submicroliter volumes.[30] nanoPOTS, the oil-air-droplet (OAD) chip, and related methods allowed significant reduction of sample preparation volumes, thus directly addressing a core challenge of delivering enough peptide material from a single cell to the mass spectrometer.[1,2,30,31] These advances in single-cell sample preparation, along with low-flow-rate separations that increased ionization efficiency and thus sensitivity, contributed to demonstrating the feasibility of using MS to quantify many proteins from individual frog embryo cells[32] and mammalian cells.[33] High-performance peptide separation is a key contributor to sensitivity improvements.[14,32,34,35] IonOpticks columns made high-performance chromatography more widely acces-

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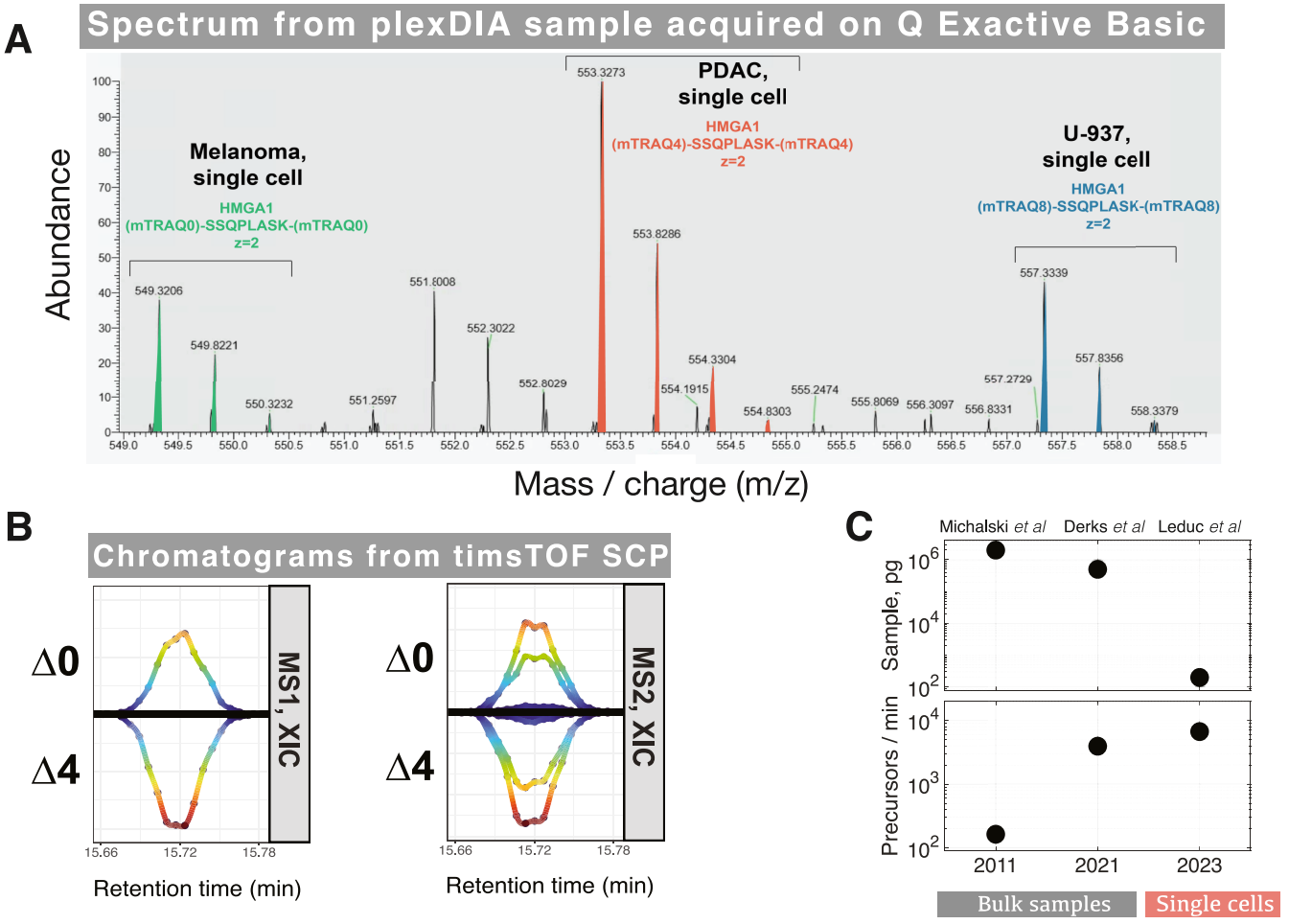


Figure 3. Spectra and throughput exemplifying the progress of MS instrumentation and methods. (A) An example MS1 spectrum showing the peptide SSQPLASK multiplexed by mTRAQ in 3 single cells analyzed by Q Exactive Basic. Data are from Derks et al.[50] (B) Extracted MS1 and MS2 chromatograms from the same peptide sequence from a plexDIA single-cell sample analyzed by slice-PASEF[59] on a timsTOF SCP system. Black circles correspond to MS scans. (C) Trends of increasing sensitivity and throughput of MS proteomics (quantified by precursors per minute from increasingly small sample amounts), illustrated by three studies representing high points for the corresponding years: Michalski et al.,[64] Derks et al.,[50] and Leduc et al.[40]

sible, thus improving the sensitivity of workflows in laboratories not specializing in peptide separations.[25]

While bulk methods have traditionally relied on detergents and chaotropic agents (e.g., urea) to lyse cells and extract proteins efficiently, we wanted to avoid using these reagents as they are usually incompatible with mass spectrometry and require removal steps that risk substantial protein loss, which is especially problematic at the single-cell scale. To avoid this, we moved toward fully MS-compatible workflows, starting with acoustic sonication in pure water, then progressing to mPOP (a freeze-heat lysis method performed directly in multiwell plates), which was shown to extract proteins as efficiently as urea-based lysis, uniformly across cellular compartments, while being far simpler to automate and scale.[24] Subsequently, many laboratories developed multiwell-based methods that provided accessible protocols, especially for label-free analysis.[36,37] Sample preparation remains an active area of development as new methods[38] seek to reduce sample losses and enable multimodal sample preparation, as recently reviewed.[2]

To minimize volumes and maximize flexibility for multiplexing, my team developed a glass-slide sample preparation method in nanoliter-volume droplets, nPOP.[39,40] This method reflected the initial ideas and aims that I had for sample preparation ca. 2014−2015, but at the time, we lacked the technology to implement it. Implementing it required the convergence of a capable piezoacoustic dispensing system, CellenONE, and a very bright and resourceful doctoral student, Andrew Leduc. nPOP reduced the reaction volumes to the nanoliter scale by performing DMSO-based lysis, digestion, and mass tag labeling in an open, flexible array of droplets on a fluorocarbon-coated slide, with humidity-controlling water droplets preventing evaporative loss. This design allows thousands of single cells to be processed in parallel while enabling quality controls, such as selecting only intact cells.[41] Rigorous controls—including isotope-labeling comparisons against urea-based lysis and negative-control droplets to monitor background contamination—confirmed that DMSO-based lysis in nPOP is as efficient and clean as established methods. The flexibility of this approach facilitates its increasing adoption and customization.[42,43] This flexibility is particularly important as new mass tags for increasing multiplexing are developed, because nPOP allows the experimental design to be adapted simply by programming the configurations of droplet dispensing.

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STRATEGIES FOR ADVANCING SINGLE-CELL PROTEOMICS

As simple preparation improved, we observed that the number of detectable peptide-like features in a single-cell run far exceeded the number of MS2 scans we could practically acquire (Figure 2).[28,44] Thus, we could not assign confident sequences to many detectable peptides within the sensitivity of MS instruments. In the absence of fragmentation spectra, peptide sequences can be assigned using match-between-runs (MBR), which is made more attractive by methods controlling its false discovery rate (FDR).[45−47] Still, we were very interested in acquiring fragmentation spectra as they increase the specificity and confidence of sequence assignment.[44]

This motivated us to think about approaches for improving sequence identification, resulting in two complementary strategies to increase the fraction of peptide-like features with confidently assigned sequences, shown in Figure 2. The first approach introduced real-time, retention-time-aligned instrument control to prioritize acquisition of biologically informative peptides, which we termed prioritized SCoPE (pSCoPE).[48,49] pSCoPE is particularly suitable when the single-cell analysis aims to focus on a set of hundreds to a couple thousand peptides of particular biological interest. Since the accumulation and overall instrument time devoted to each peptide can be individually tailored, the sensitivity for peptides of interest can be substantially increased. This can be particularly beneficial for peptides with post-translational modifications. In our experiments, this extended dynamic range by an order of magnitude compared to controlled shotgun analysis.[48]

The second approach was data-independent acquisition (DIA), which parallelizes the fragmentation of peptides, and multiplexed DIA (plexDIA), which parallelizes both peptide fragmentation and sample analysis.[50,51] Thus, plexDIA enabled multiple single cells to be analyzed within a single run while coisolating and fragmenting multiple peptides per MS2 scan.[52,53] The parallel precursor accumulation and fragmentation during DIA allows longer accumulation times, thus increasing sensitivity for single-cell protein analysis.[44]

Parallelizing the isolation and fragmentation of precursors by DIA or plexDIA allowed for shortening peptide separation times and increasing throughput for proteomic analysis,[54,55] especially for single cells.[56] These throughput gains are particularly significant when using chromatographic approaches with reduced overhead, such as Evosep or dual-column setups.[56,57] Combining dual-column setups with short-gradient plexDIA has already supported analyzing over 1000 single-cell-level samples per day[58] and holds the potential to further increase throughput.

The advances in sample preparation, peptide separation, and data acquisition allowed us to make highly quantitative proteomic measurements even when using older instruments such as the Q Exactive Classic (Figure 3A). These measurements showed strong agreement between spiked-in and measured protein and peptide levels[48] and excellent agreement between MS1- and MS2-level quantification by plexDIA.[50] These capabilities were substantially enhanced by parallel advances in instrumentation from Bruker (the timsTOF line)[59,60] and Thermo Fisher (Orbitrap Astral platforms),[61−63] which have further extended achievable depth, accuracy, and throughput as recently reviewed.[2] The increased scanning speed of these instruments allows frequent sampling of elution peaks (Figure 3B). This speed, combined with more efficient ion transfer and detection and with new data acquisition and interpretation methods, has increased proteomic throughput while decreasing the required sample amounts by orders of magnitude (Figure 3C).

INCREASING THROUGHPUT VIA PARALLELIZATION

As single-cell proteomics matured, early skepticism gave way to enthusiasm, fueled by frameworks such as plexDIA that offered a path to scale throughput without sacrificing—and potentially improving—proteome depth and quantitative accuracy.[51−53] This prospect attracted the interest of Convergent Research and the philanthropy of Eric and Wendy Schmidt and Griffin Catalyst, who encouraged us to accelerate multiplexed mass spectrometry proteomics through a focused research organization. This led to cofounding the Parallel Squared Technology Institute (PTI) with my former doctoral students Harrison Specht and Aleksandra Petelski. We established PTI not only to advance single-cell proteomics, but also to translate its technological capabilities into biomedical research, with a research emphasis on neurodegenerative diseases[65] and functional and conformational protein modifications.

Many biomedical research projects demand the analysis of thousands of single cells.[53] Consider, for example, investigating proteome changes in single cells from post-mortem brain samples from Alzheimer's disease patients. Analyzing each brain region requires sampling on the order of a thousand single cells to meaningfully represent its major cell types. Thus, sampling four brain regions from a modest cohort of 30 Alzheimer's disease patients would require analyzing over 100,000 single cells.

To enable studies at this scale and beyond, the PTI team has developed next-generation strategies for multiplexing in both the mass and time domains, through reagents for plexDIA, such as PSMtags,[58] methods for parallelizing sample analysis in time, such as timePlex,[66] and computational frameworks for joint modeling of mass spectra (JMod)[67] to facilitate the interpretation of highly parallelized mass spectra. These approaches and ongoing developments aim to bring single-cell proteomics throughput to many thousands of cells analyzed per day while preserving deep coverage and quantitative accuracy. Together, these advances point toward a future in which single-cell proteomics and functional measurements become practical for large-scale biomedical discovery, including the training of AI models, and may ultimately lead to new biological and biophysical principles.[2,68]

WHY MASS SPECTROMETRY

Single-cell proteomics has developed alongside single-cell RNA sequencing, a field that emerged around 2010 and attracted tens of billions of dollars of investment, orders of magnitude more than single-cell protein analysis. This disparity raises two questions worth discussing. First, what might the field look like today had mass-spectrometry-based single-cell proteomics been introduced alongside sequencing in 2010, developing in parallel rather than playing catch-up? Second, given how mature and accessible single-cell DNA sequencing has become, is mass spectrometry still necessary?

I believe the answer is unambiguously yes, and not merely because of my own investment in the technology. The proteome and transcriptome are fundamentally different in scale: a typical cell contains fewer than a million mRNA molecules but billions of protein molecules. In terms of cost per molecule counted,

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mass spectrometry proves to be the more economical method for counting molecules of any kind, not only proteins, compared with even the most highly optimized short-read sequencing.[69,70]

More fundamentally, mass spectrometry measures far more than molecular abundance; it can measure protein modifications and rates of protein synthesis and degradation, which are essential to understanding cellular biology. For example, feeding mice with food containing heavy isotopes allowed us to measure protein synthesis and degradation dynamics inside single cells within a living mammal[71]—an experiment no other current technology can perform. Single-cell RNA sequencing can perform pulse-labeling of transcripts with 4-thiouridine (4sU), but it has not allowed for quantitative in vivo measurements. Thus, mass spectrometry is unique in being able to capture in vivo protein dynamics at single-cell resolution in intact tissue.

Furthermore, mass spectrometry has allowed measuring post-translational modifications in single cells[72,73] and even proteolytic cleavages in single macrophage cells, thus revealing differences in signaling mechanisms associated with functional polarization.[48] Future methodological developments will extend more of the MS analytical repertoire to single cells. For example, powerful protein footprinting methods that allow measuring changes in surface accessibility in vivo[74] have the potential to be extended to single cells from mammalian tissues.[75]

PROTEIN REGULATION IN SINGLE CELLS

Single-cell proteomics has opened the possibility of inferring post-transcriptional regulation from single-cell proteogenomic data. This requires distinguishing genuine biological differences between mRNA and protein levels from artifacts of measurement noise, missing data, and computational integration.[68,76] Because transcript and protein abundances are typically collected using distinct technologies with different and largely nonoverlapping technical biases, apparent discrepancies between the two modalities can arise from technical sources—such as ratio compression in isobaric labeling, dropout in scRNA-seq, or errors introduced during cross-modality cluster alignment—rather than from true regulation of protein synthesis or degradation. Thus, we focused on developing methods, such as BayesPG, for quantifying noise, reliability, and biological regulation.[77] Similarly, other laboratories have used parallel RNA and protein measurements in single cells to understand differences in their regulation, such as in the context of hematopoiesis.[78]

Inferring biological regulation is particularly powerful when the rates of protein synthesis and degradation can be estimated based on metabolic pulse labeling with heavy isotopes as discussed above. Because histone proteins are degraded very slowly, the incorporation of heavy isotopes into histones let us count how many times each cell has divided and thereby estimate cell-type-specific growth rates. Such analysis revealed a simple and, to us, striking principle: roughly half of the variation in protein abundance among nonproliferative cells is determined by protein degradation rates, and the degree to which degradation controls abundance is itself set by the cell division rate—a relationship that follows naturally from a basic biophysical model of protein clearance through growth, division, and degradation.[71] As cells proliferate faster, degradation's contribution to abundance diminishes; but most cells in an organism persist for a long time, and for those cells, protein degradation strongly determines protein abundance and cellular phenotype. Clearance rates regulate differences across single cells from the same cell type,[71] shape spatial protein gradients associated with liver zonation,[67] and set the ratios among proteoforms encoded by the same gene, including splicing isoforms.[79]

PROTEIN COVARIATION ACROSS SINGLE CELLS: A WINDOW TO CELLULAR FUNCTIONS

I have always been inspired by the potential of single-cell proteomics to reveal biological functions and to open new perspectives on how cellular processes are organized.[80,81] Single-cell proteomics may transform natural cell-to-cell variation into a rich source of "natural experiments" for uncovering biological regulation. Covariation of protein abundances across single cells may reveal cell-type- and cell-state-specific protein complexes and functional relationships, while joint single-cell analysis of proteins and transcripts can expose post-transcriptional regulatory interactions invisible to RNA measurements alone. This perspective shifts the focus from heterogeneity across cells to coherent patterns of covariation.[81] It reframes the cataloging of differences as a quest for simplicity, order, and principles.[2]

Remarkably, protein covariation across single cells often reveals functionally coherent molecular organization to a far greater degree than the coordination at the level of mRNA covariation across the same populations of single cells.[17,27,82,83] Indeed, protein−protein covariation is much stronger and more coherent than the corresponding mRNA−mRNA covariation measured with the deepest available sequencing, as shown in Figure 4.[71] The protein covariation structure is stronger because

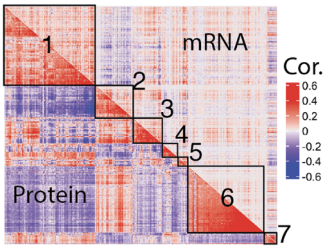


Figure 4. Protein−protein correlations within a cell type are more functionally coherent than the corresponding mRNA−mRNA correlations. The upper triangular matrix displays pairwise Pearson correlations between mRNAs across basal cells from mouse trachea. The lower triangular matrix displays the corresponding pairwise correlations between proteins across single basal cells. The coherent clusters of protein correlations display strong enrichment for biological functions as follows: (1) cytoskeleton organization, (2) xenobiotic metabolism, (3) carbohydrate metabolism, (4) mesenchymal transition, (5) olefinic compound metabolic process, (6) aerobic respiration, and (7) eukaryotic translation. This figure is adapted from Leduc et al.[71] Copyright 2025 Nikolai Slavov.

of (1) lower measurement noise and (2) much of protein covariation is directly attributable to coordinated regulation of protein synthesis and degradation. These factors make protein measurements substantially more biologically interpretable.[71] The interpretability of single-cell protein measurements is bolstered by a recent analysis indicating that the single-cell variation in protein abundances measured by MS is dominated by biological factors, while technical noise is relatively small.[84]

We have applied this protein-covariation framework to many biological questions to identify functionally coordinated protein changes. For example, this approach revealed dynamic reorganization of the proteome of cancer cells undergoing epithelial-to-mesenchymal transition.[82] In another project, we

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J. Am. Soc. Mass Spectrom. XXXX, XXX, XXX−XXX

identified cell-type-specific differences in the coordination of functional protein modules with the cell division cycle of immune cells and melanoma cells.[39] The approach also helped identify cell-type-specific functional coordination of proteins in human immune cells.[85] Further, we used single-cell protein covariation to identify putative regulators of protein transport to the nucleus.[86] Importantly, the regulatory roles of these proteins were later tested and confirmed through targeted perturbation experiments, thus providing compelling evidence for the feasibility of inferring regulatory interactions from joint protein distributions across single cells.[86]

DEVELOPMENTAL BIOLOGY

Developmental biology particularly needs single-cell proteomics because many of the earliest regulatory events are post-transcriptional.[87] In oocytes and early embryos, maternally deposited proteins and the regulation of their synthesis, localization, modification, and degradation can decouple protein states from transcript abundance. Indeed, single-cell proteomics has long been used to study frog development[32] and even single human oocytes.[88] We investigated the effects of maternal age on the oocyte proteome and found decreased abundance of proteostasis and meiosis proteins in oocytes from women of advanced maternal age.[89] In another project, we investigated the earliest breaking of symmetry between sister blastomeres in mouse and human two-cell embryos. Circumstantial evidence had long suggested that the two cells at this stage are already different, but roughly a dozen high-profile RNA sequencing studies had failed to identify the molecular basis. Protein measurements told a different story: hundreds of proteins were robustly and reproducibly different between sister blastomeres, allowing us to define α and β blastomere identities and to show that this asymmetry is not merely molecular noise but reflects functional specialization that predicts distinct developmental fates.[29]

UNDERSTANDING NEURODEGENERATION WITH SINGLE-CELL PROTEOMICS

Single-cell proteomics is beginning to answer biological questions that were previously inaccessible. As discussed here, it opens new perspectives that can be particularly valuable and powerful in the context of neurodegeneration. Indeed, protein pathologies play central roles in neurodegeneration,[65,74,90,91] including alterations in their translation,[92] post-translational modifications,[93,94] and degradation.[79,95−97] Since neurons are nonproliferative, long-lived cells, protein stability is a major factor determining protein abundance,[79] and reduced clearance of damaged proteins may directly contribute to neurodegeneration.[65] Studying these pathological mechanisms requires direct protein measurements. However, population-average measurements of diverse cell types face major challenges in deciphering the underlying molecular mechanisms, which may be overcome by scalable, deep, and quantitative protein analysis in single cells.[14] This possibility motivates PTI's focus on single-cell proteomic analysis of neurodegenerative disease, including both Alzheimer's and Parkinson's diseases. Toward this goal, the proteomes of single cells can be sampled by patch-clamping,[98−100] laser capture microdissection,[101] or dissociation of tissue samples into single cells, which requires careful, deliberate optimization to ensure that the analyzed cells are genuine cells rather than debris.[41] These approaches allow identifying proteins that are differentially abundant across different cell types, including neuronal subtypes such as excitatory glutamatergic and inhibitory GABAergic neurons.

Crucially, single-cell proteomics allows for new kinds of analysis that can overcome major challenges to previous efforts. Specifically, the single-cell resolution enables staging the disease progression of each individual cell.[80,102] Thus, instead of assuming that all cells from a given patient, characterized by a single pathology score and disease stage, are at the same point in disease progression, the single cells may be ordered along a trajectory of disease progression, thus revealing disease dynamics. Furthermore, single-cell proteomics can reduce confounding from genetic and lifestyle differences inherent to bulk studies. Because cells from the same patient largely share their genetic background and environmental history, variation among them can reveal disease dynamics and regulation within a common context. For example, we can identify disease-associated proteins that covary within a patient—such as α-synuclein and specific ATPases—and extend these associations to functionally coherent protein sets that generate testable hypotheses.[81] More broadly, analyzing cells within patients allows us to condition on many between-patient confounders and instead identify cells at different stages of disease progression,[2] including cells undergoing programmed cell death.[103]

BUILDING A COMMUNITY AND OUTLOOK

None of this work would have been possible without a community of collaborators who share both the scientific goals and a commitment to holding one another to a high standard. The emergence of single-cell proteomics was propelled by complementary advances developed by different groups, such as miniaturized sample preparation and sensitive chromatography introduced by the Kelly group,[33,34] critical data reanalysis and open-source packages by the Gatto group,[18] initial guidelines formulated with community leaders, including John Yates III and Ruedi Aebersold,[104] free software from several groups for sensitive sequence assignment to mass spectra,[23,46,47,59,105−107] and early application of sensitive MS instruments by the Mann, Olsen, and Mechtler groups.[60−63] The field became convincing when these independent advances converged and repeatedly demonstrated that single-cell proteomes could be measured quantitatively and reproducibly.

In 2018, we organized the first single-cell proteomics conference at Northeastern University.[108] The intention from the outset was to build a community: people who know each other, span diverse fields while sharing common objectives,[109] and can push one another, in a supportively critical way, toward the field's full potential. From the beginning, we have emphasized open sharing not just of raw data but of metadata, which proved invaluable during the field's early, more skeptical years—allowing others to reproduce our analyses and judge the evidence for themselves. Our joint efforts have established community-developed guidelines to help newcomers adopt these methods[104] and continue to build common resources and infrastructure, such as cloud-based data analysis[106] and a community-oriented single-cell proteomics portal at PRIDE.

Participating in the development of single-cell proteomics by mass spectrometry has been a wonderful experience illustrating the power of proof-of-principle demonstrations, creative and rigorous approaches, interdisciplinary collaboration, and the remarkable upward thrust of technological development (Figure 3). It has opened new perspectives on biological systems that other methods cannot access. I believe our community is at a

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https://doi.org/10.1021/jasms.6c00362
J. Am. Soc. Mass Spectrom. XXXX, XXX, XXX−XXX

particularly exciting juncture: our collective efforts can meaningfully enlarge the vistas this technology has opened, profoundly deepen our fundamental understanding of biological systems, and ultimately contribute to developing more effective therapies.

■ ASSOCIATED CONTENT

Related Articles

This article is an invited perspective for the special issue of the Journal of the American Society of Mass Spectrometry (JASMS) celebrating the 2026 Biemann Medal recognizing the development of single-cell proteomics by mass spectrometry. A recording of the corresponding lecture is available at: https://youtu.be/z0dIhG4C3qI?si=lKQdAcYmo0-dy2O4.

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Funding

N.S. is supported by NIGMS award R35GM148218, NIA award R01AG092460, and NSF award 2534124.

Notes

The author declares the following competing financial interest(s): N.S. is a founding director and CEO of Parallel Squared Technology Institute, which is a non-profit research institute. The author is co-inventor of patents for single-cell proteomics methods and has collaborative research agreements with Bruker and Thermo Fisher, who are manufacturers of the mass spectrometry instrumentation.

■ ACKNOWLEDGMENTS

I am deeply grateful to everyone who has contributed to this work over the years, including members of my research teams at Northeastern University and at the Parallel Squared Technology Institute, our collaborators, and many others. Our joint efforts not only developed single-cell proteomics and used it to open new windows onto biology but also made the process highly rewarding and genuinely enjoyable. I thank B. Karger for critical comments on this perspective.

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